Consider the gas-phase reaction below.

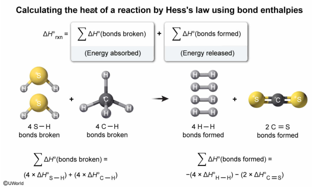
$$2\ H_2S(g) + CH_4(g) \rightarrow 4\ H_2(g) + CS_2(g)$$

Which expression can be used to calculate the heat of reaction ($\Delta H°_{rxn}$) for the reaction shown using bond enthalpies ($\Delta H°_{bond}$)?

- A. $\Delta H°_{rxn} = (2 \times \Delta H°_{S-H}) + \Delta H°_{C-H} - (4 \times \Delta H°_{H-H}) - \Delta H°_{C=S}$ [8%]
- B. $\Delta H°_{rxn} = (4 \times \Delta H°_{S-H}) + (4 \times \Delta H°_{C-H}) - (4 \times \Delta H°_{H-H}) - (2 \times \Delta H°_{C=S})$ [36%]
- C. $\Delta H°_{rxn} = (4 \times \Delta H°_{H-H}) + (2 \times \Delta H°_{C=S}) - (4 \times \Delta H°_{S-H}) - (4 \times \Delta H°_{C-H})$ [52%]
- D. $\Delta H°_{rxn} = (4 \times \Delta H°_{H-H}) + (2 \times \Delta H°_{C=S}) + (4 \times \Delta H°_{S-H}) + (4 \times \Delta H°_{C-H})$ [3%]

Correct
37% answered correctly

Explanation:



Bond enthalpy $\Delta H°$ (bond dissociation energy) is the amount of energy needed to break 1 mole of a bond between two atoms in the **gas phase** at a temperature of 298 K. Forming a bond is an exothermic process (**$\Delta H°_{\text{bonds formed}}$ < 0**) whereas **breaking a bond** is an endothermic process (**$\Delta H°_{\text{bonds broken}}$ > 0**). Consequently, the enthalpies associated with breaking or forming a given type of bond are equal in magnitude but opposite in sign.

A chemical reaction can be viewed as a series of steps in which reactant bonds are broken and product bonds are formed. The overall **enthalpy change of a reaction $\Delta H°_{rxn}$** measures the heat released or consumed (ie, $\Delta H°_{rxn} < 0$ or $\Delta H°_{rxn} > 0$, respectively) by 1 mole of reaction. Hess's law states that if a reaction occurs in a series of intermediate steps, $\Delta H°_{rxn}$ is equal to the **sum** of the enthalpy changes for each step.

According to Hess's law, $\Delta H°_{rxn}$ can be calculated by

$$\Delta H°_{rxn} = \sum \Delta H°_{(\text{bonds broken})} + \sum \Delta H°_{(\text{bonds formed})}$$

In the reaction given, 4 moles of S–H bonds and 4 moles of C–H bonds are broken:

$$\sum \Delta H°_{(\text{bonds broken})} = (4 \times \Delta H°_{S-H}) + (4 \times \Delta H°_{C-H})$$

Furthermore, 4 moles of H–H bonds and 2 moles of C=S bonds are formed during the reaction:

$$\sum \Delta H°_{(\text{bonds formed})} = -(4 \times \Delta H°_{H-H}) - (2 \times \Delta H°_{C=S})$$

Therefore, $\Delta H°_{rxn}$ can be expressed mathematically as the sum of the corresponding enthalpy changes for each step involving bond breakage (enthalpy is positive) and bond formation (enthalpy is negative):

$$\Delta H°_{rxn} = (4 \times \Delta H°_{S-H}) + (4 \times \Delta H°_{C-H}) - (4 \times \Delta H°_{H-H}) - (2 \times \Delta H°_{C=S})$$

(Choice A) This expression does not include the correct number of moles of S–H and C–H bonds broken and C=S bonds formed during the reaction.

(Choices C and D) Bond breakage *absorbs* energy (+$\Delta H°$), and bond formation *releases* energy (−$\Delta H°$)

Educational objective:
Bond enthalpy ($\Delta H°$) is the amount of energy needed to break 1 mole of a bond between two atoms in the gas phase. Energy is absorbed ($\Delta H° > 0$) when a bond is broken, and energy is released ($\Delta H° < 0$) when a bond is formed. Hess's law states that the standard enthalpy change ($\Delta H°_{rxn}$) of a reaction is equal to the sum of the enthalpy changes for each step involved in bond breakage and formation during a reaction.

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Thermodynamics, Kinetics & Gas Laws